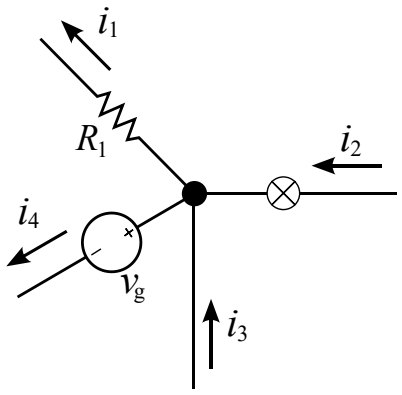
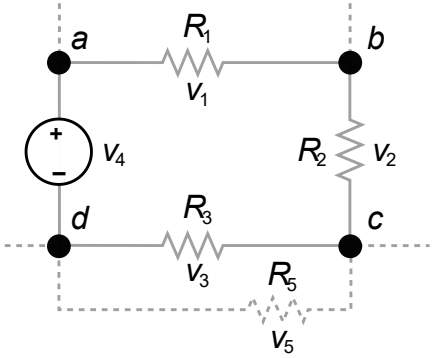
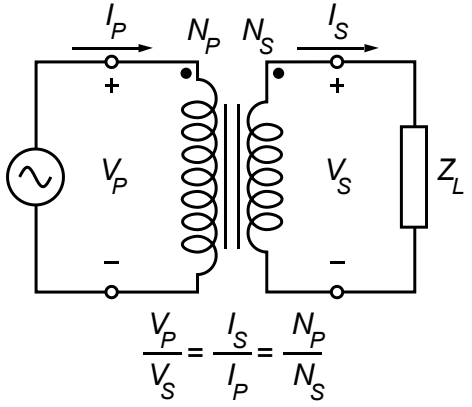
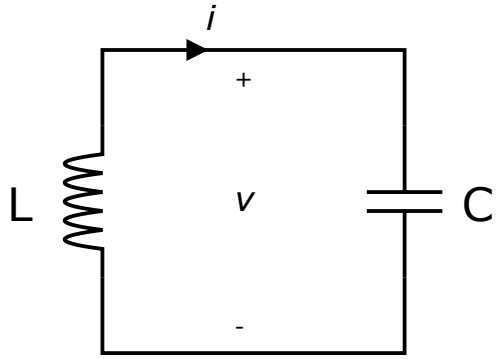
Hochschule Kaiserslautern
University of Applied Sciences

Introduction to Electrical Engineering









Overview

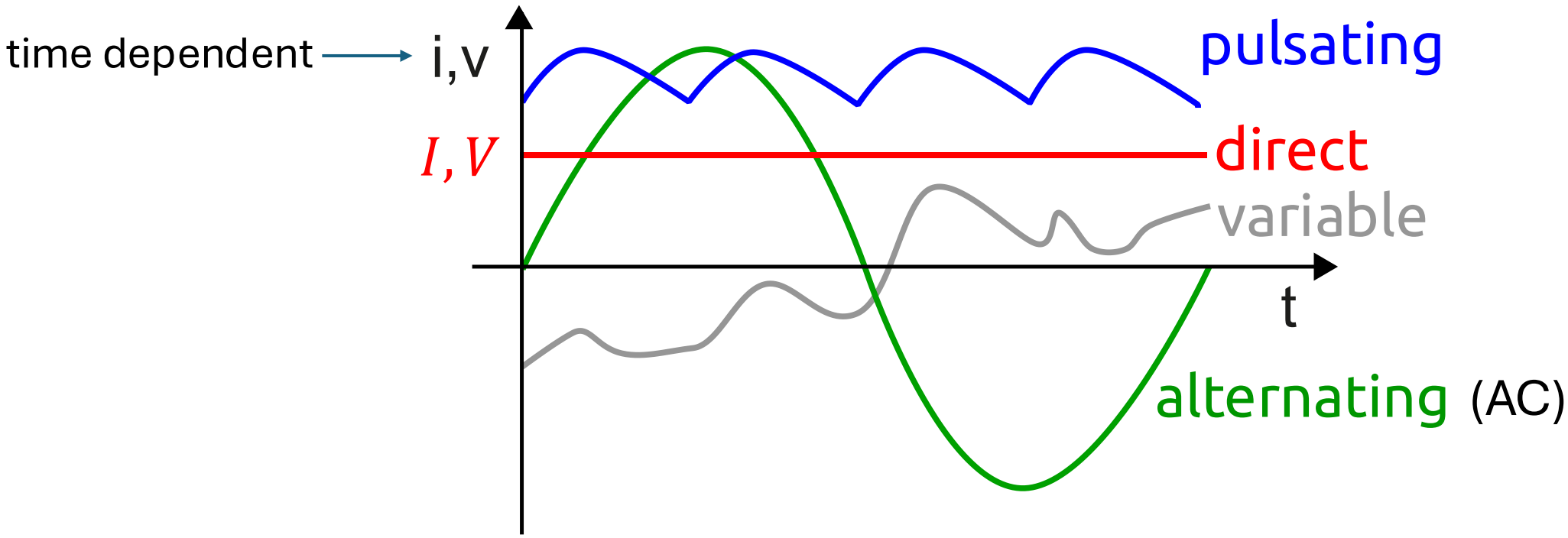
1. Physical Principles and Basic Concepts
2. **Direct Current Circuits**
3. Electric and Magnetic Fields
4. Alternating Current Technology
5. Wrap-Up and Outlook

Direct Current Circuits

- Reference Directions
- Kirchhoff‘s Current Law
- Kirchhoff‘s Voltage Law
- Series Circuit
- Parallel Circuit
- Practical Voltage & Current Sources
- Network Analysis
- Efficiency & Power/Impedance Matching
- Voltage & Current Measurement

Direct Current (DC)

one-directional (typically constant) flow of electric charge



Circuit Calculations

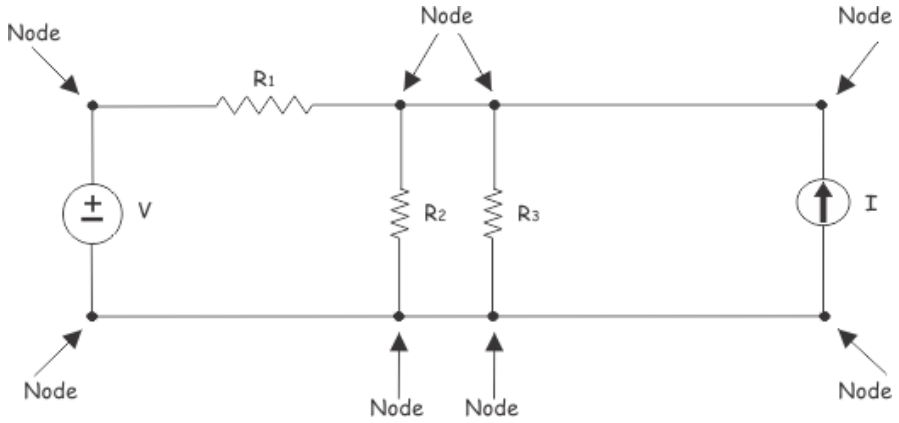
goal: calculate currents and voltages

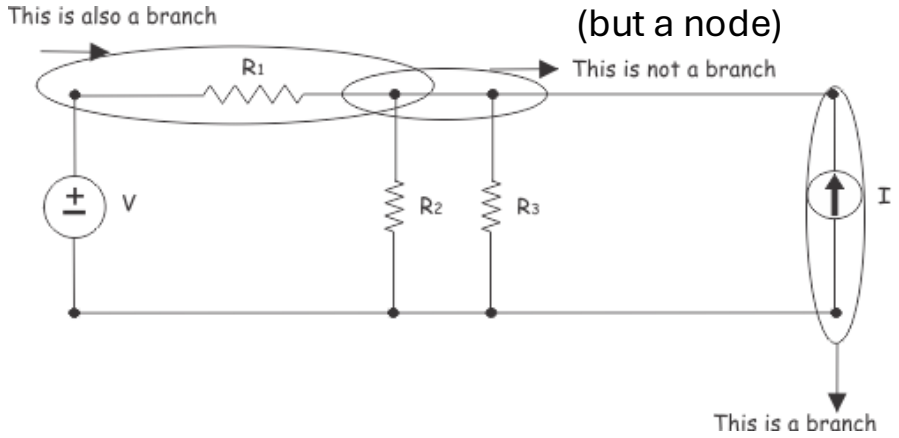
network: interconnection of electrical components/elements (often large-scale)

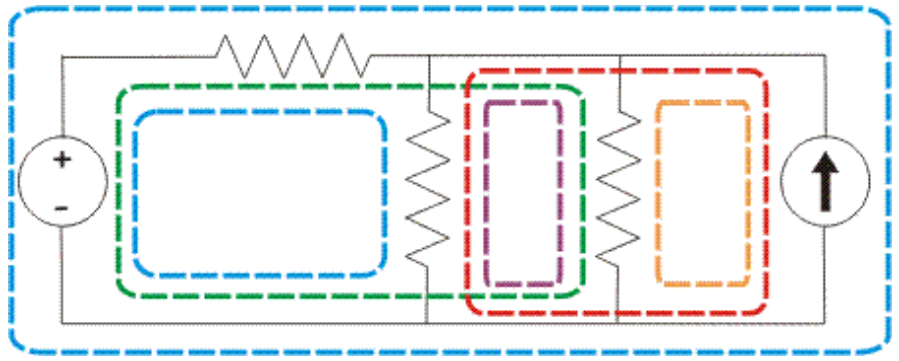
circuit: closed-loop network (rather small)

fundamental concepts:

- nodes (junctions)
- branches
- loops







Short and Open Circuits

short circuit: circuit element with resistance approaching zero

open circuit: circuit element with resistance approaching infinity

Reference Directions

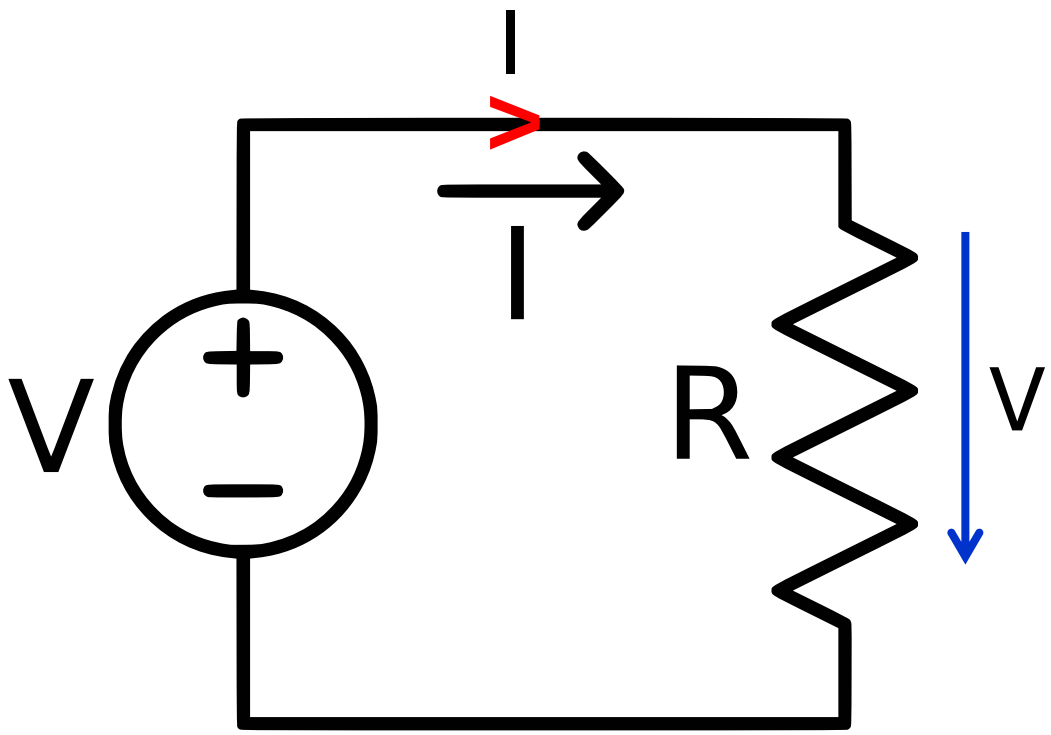
before calculation of voltages and currents:

arbitrary choices

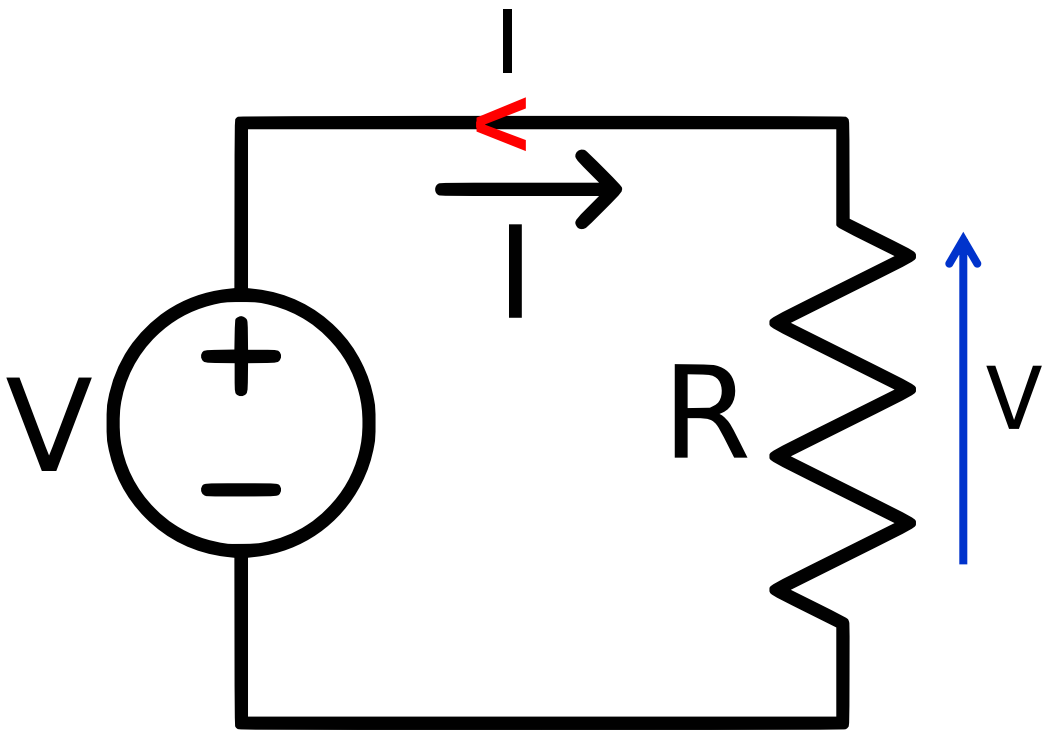
- define positive current directions
- define voltage directions + to −

after calculation:

negative value → actual direction opposite of chosen reference direction



also correct:



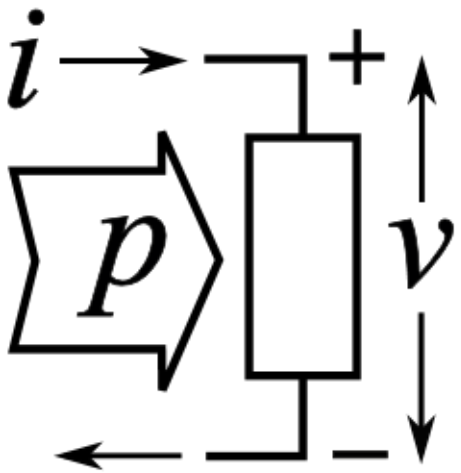
Will just result in negative values for I and V!

Sign Conventions

passive sign convention:

reference direction of current points into positive reference terminal of voltage

- $P > 0$ for power flowing into component (passive component)
- $P < 0$ for active component

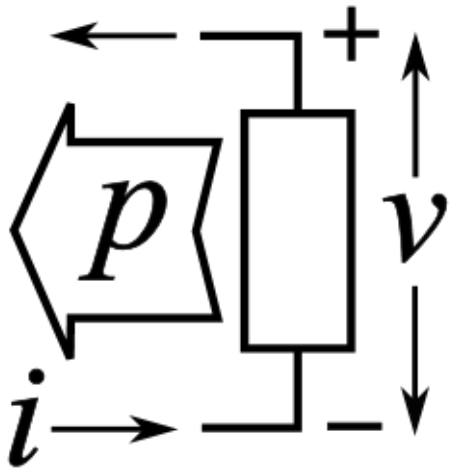


more common

active sign convention:

reference direction of current points into negative reference terminal of voltage

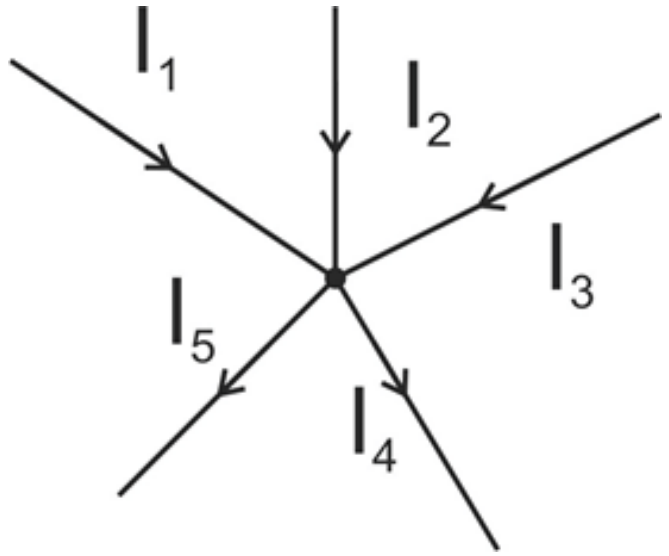
- $P > 0$ for power flowing out of component (active component)
- $P < 0$ for passive component



Kirchhoff‘s Current Law (KCL)

sum of all currents entering a junction (node) equals sum of all currents leaving that junction (conservation of charge)

$$\sum_{k=1}^{n} I_k = 0$$

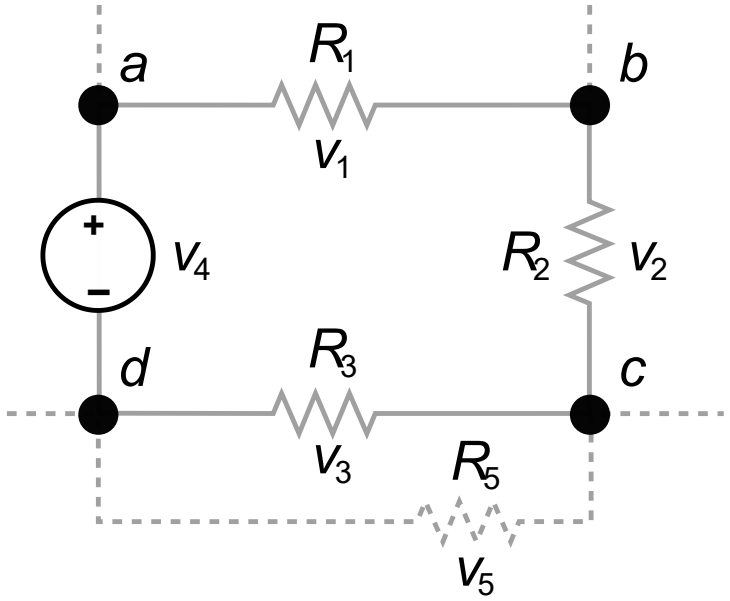


$$I_1 + I_2 + I_3 = I_4 + I_5$$

Kirchhoff‘s Voltage Law (KVL)

sum of all voltage drops around any closed loop equals to zero (conservation of energy)

$$\sum_{k=1}^{n} V_k = 0$$

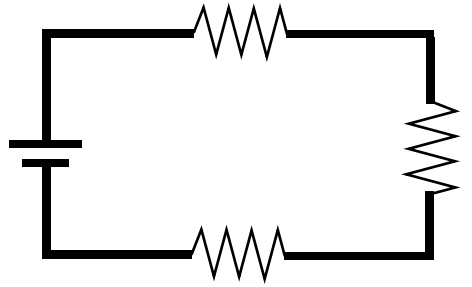


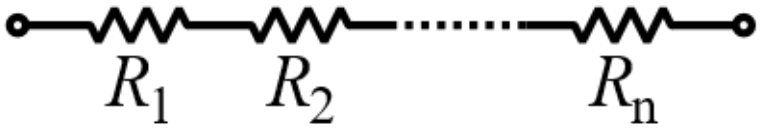
$$V_4 = V_1 + V_2 + V_3$$

Series Circuit

- sum of potential drops around a closed loop equals supply voltage (KVL)
- same current through all components (only one path)

→ $V = V_1 + V_2 + V_3 = IR_1 + IR_2 + IR_3 = IR$





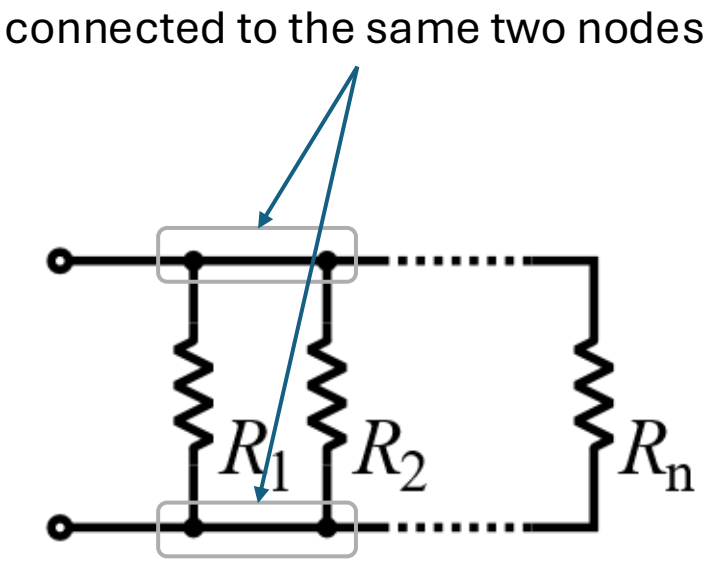
$$R = \sum_{k=1}^{n} R_k$$

equivalent resistance:
exhibits same $i - v$ characteristics as original network at considered terminals

voltage divider rule: $V_k = V \frac{R_k}{R}$

for two resistors: $\frac{V_1}{V_2} = \frac{R_1}{R_2}$

Parallel Circuit



- sum of branch currents equals current from source (KCL)
- same voltage across each branch (KVL)

$$\rightarrow I = I_1 + I_2 + \cdots + I_n = \frac{V}{R_1} + \frac{V}{R_2} + \cdots + \frac{V}{R_n} = \frac{V}{R}$$

$$R = \left(\sum_{k=1}^{n} \frac{1}{R_k} \right)^{-1} \qquad G = \sum_{k=1}^{n} G_k$$

current divider rule: $I_k = I \frac{R}{R_k}$

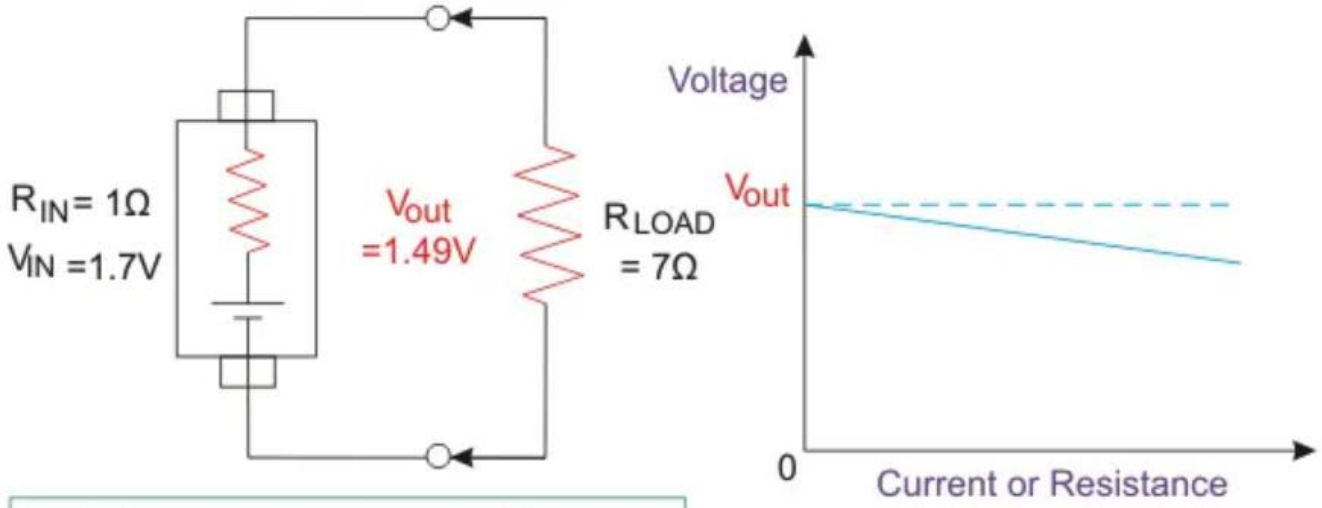
for two resistors: $\frac{I_1}{I_2} = \frac{R_2}{R_1}$; $R = \frac{R_1 R_2}{R_1 + R_2}$; $I_1 = I \frac{R_2}{R_1 + R_2}$

Practical Voltage Source

equivalent circuit
of real component

ideal voltage source in series with internal resistance

→ voltage dependent on load

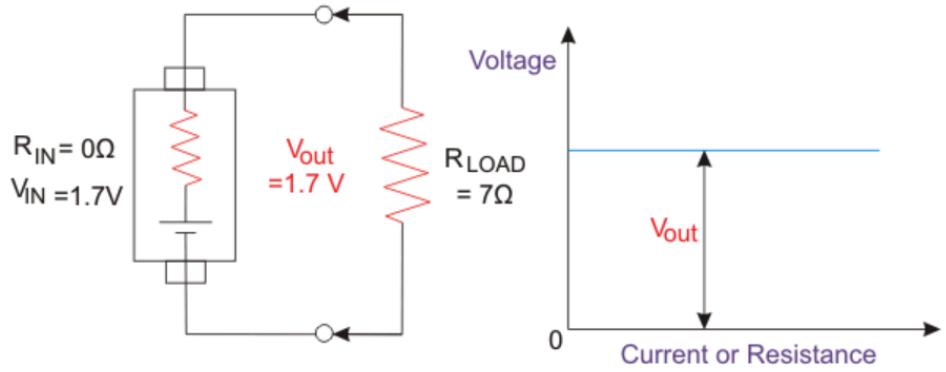


$I=V/(R_{LOAD}+R_{IN})=1.7/8=0.2125A$

R_{IN} Source Voltage = I X R_{IN}=0.2125V

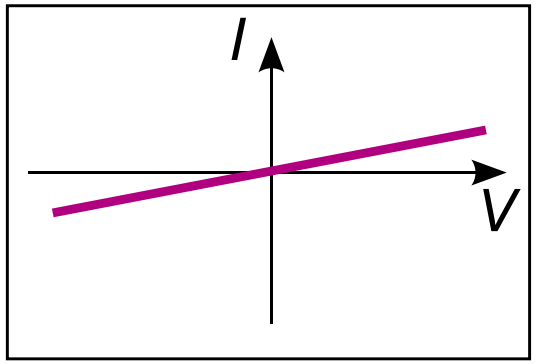
So, V_{out} =1.7 - 0.2125=1.49V

ideal voltage source:

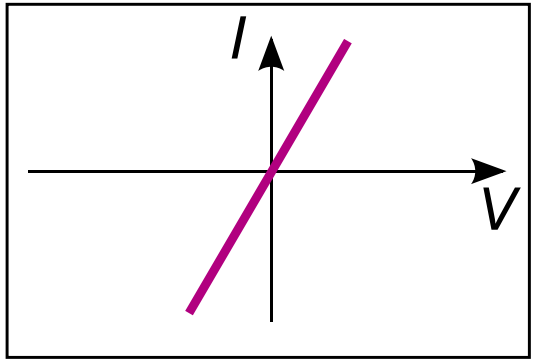


I-V Curves

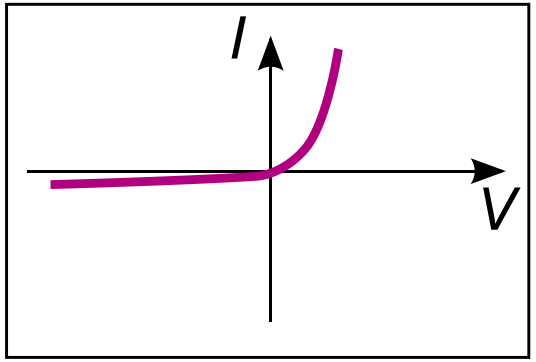
Large resistance



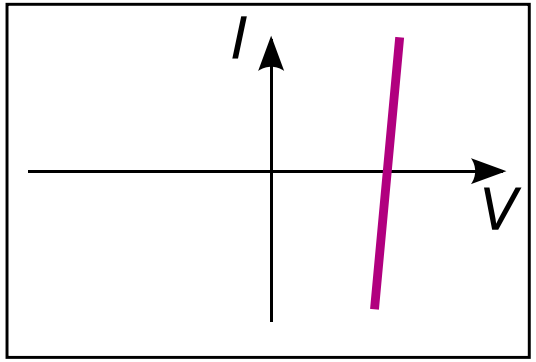
Small resistance



Diode



Battery



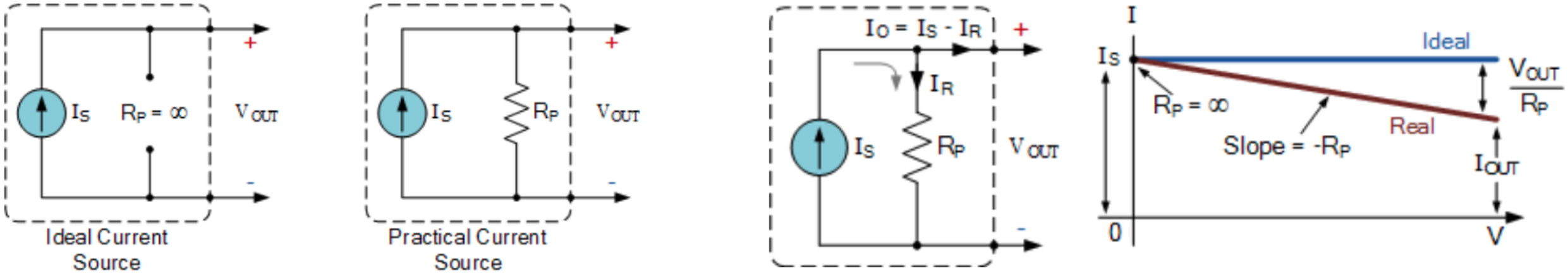
nonlinear

(considerations here only apply to linearized region around operating point)

Practical Current Source

internal resistance in parallel with ideal current source

→ current dependent on load



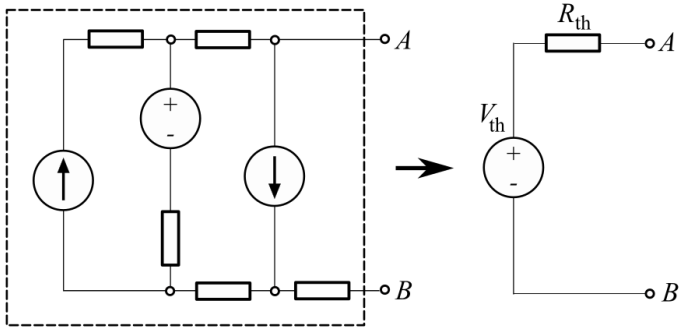
Several Independent Sources

simplification by means of superposition principle in linear circuits:

- turn on one source at a time (others turned off: replace voltage sources with short circuits and current sources with open circuits)
- find its contribution to the terminal voltage/current
- add all contributions

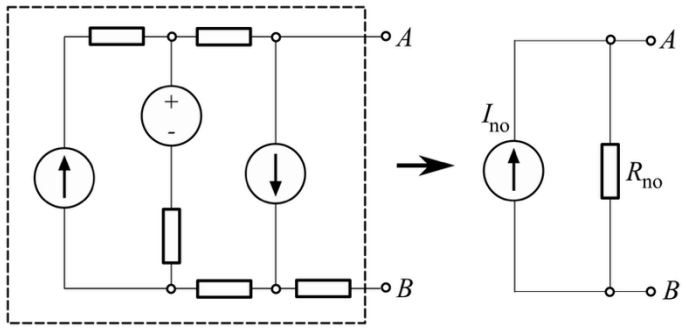
additionally, single “turn off all” reduction to get R_{th}

Thévenin‘s theorem:

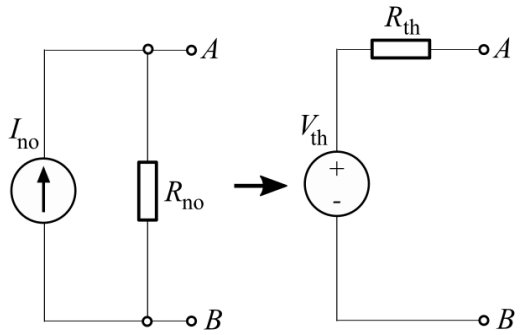


any linear circuit seen from two terminals

Norton‘s theorem:



Norton-Thévenin conversion:



$$R_{\mathrm{th}} = R_{\mathrm{no}}$$
$$I_{\mathrm{no}} = V_{\mathrm{th}}/R_{\mathrm{th}}$$

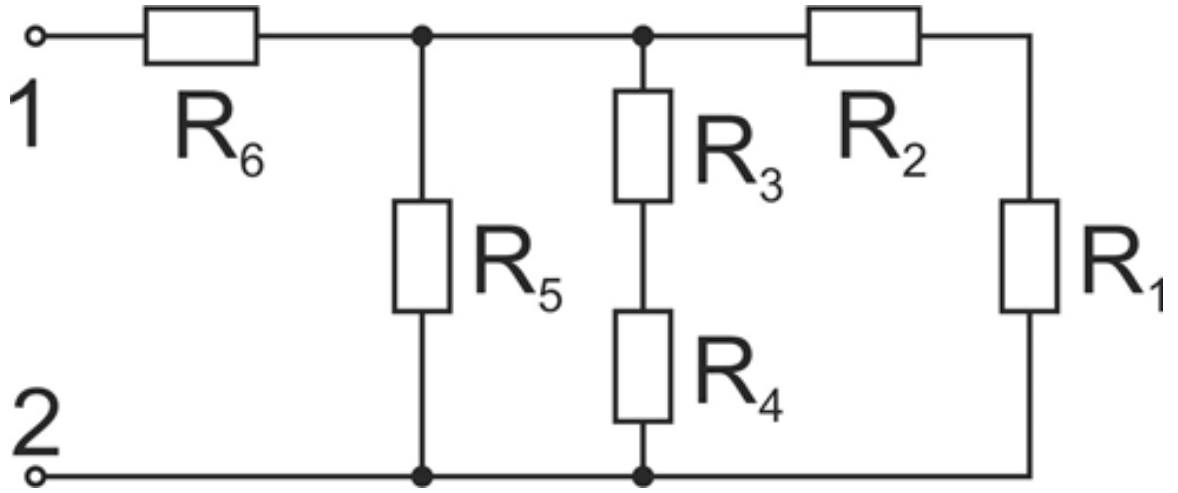
Circuit Analysis

approach for mixed circuits:

- reduce series groups (same current path): add resistances
- reduce parallel groups (same two nodes): add conductances
- redraw after each combine, repeat

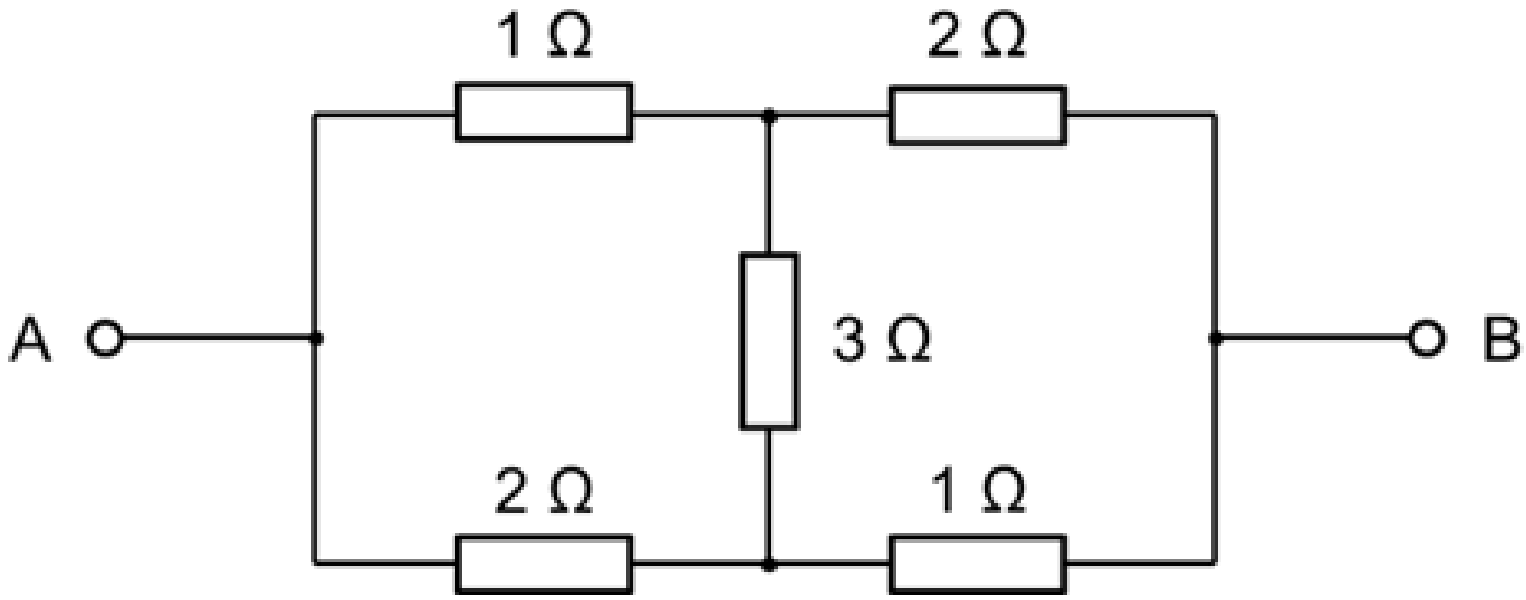
example: total resistance between 1 and 2

$$R_{1,2} = R_1 + R_2, \qquad R_{3,4} = R_3 + R_4$$
$$G_{1,2,3,4} = G_{1,2} + G_{3,4}$$
$$G_{1,2,3,4,5} = G_{1,2,3,4} + G_5$$
$$R = R_{1,2,3,4,5} + R_6$$



Bridges

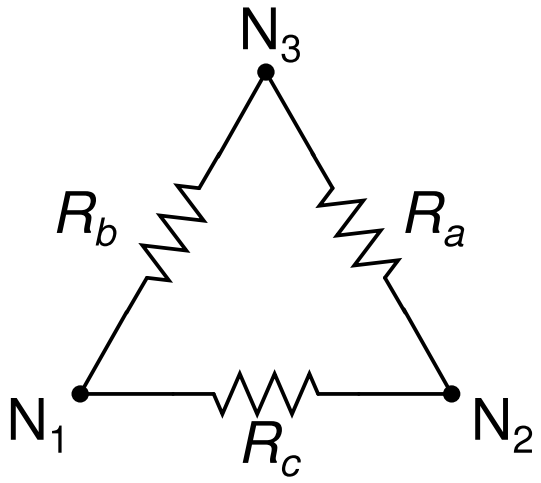
How to solve this one?

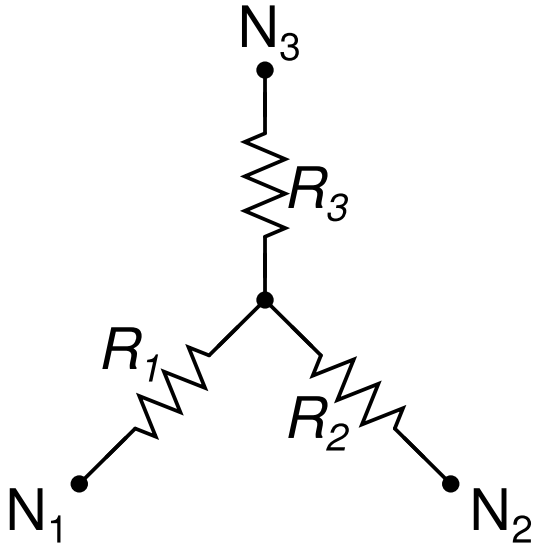


→ convert the bridge with Y-Δ transform

(simplification by using three-terminal equivalent networks)

Y-Δ Transform





idea of Y-Δ equivalence:
define same effective resistance between corresponding terminals

e.g., for N_1 and N_2:

$$R_1 + R_2 = \left(\frac{1}{R_c} + \frac{1}{R_a + R_b}\right)^{-1}$$

transformation from Δ to Y:

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$

transformation from Y to Δ:

$$R_a = R_2 + R_3 + \frac{R_2 R_3}{R_1}$$

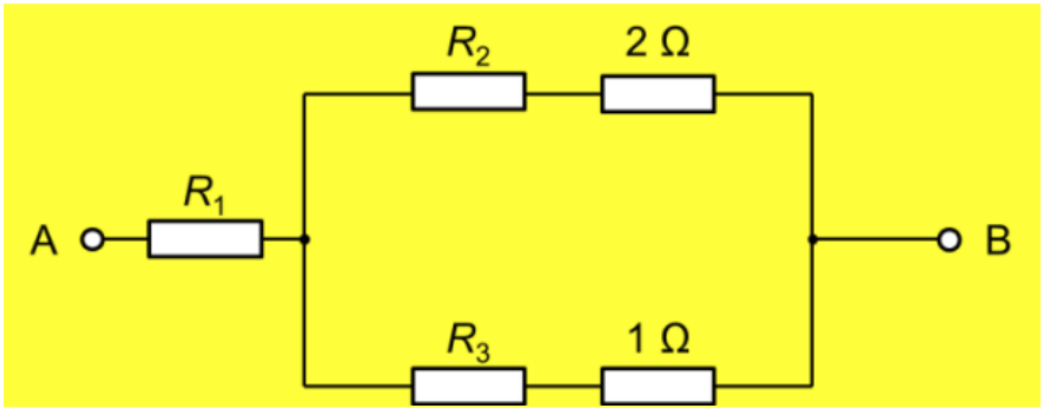
$$R_b = R_1 + R_3 + \frac{R_1 R_3}{R_2}$$

$$R_c = R_1 + R_2 + \frac{R_1 R_2}{R_3}$$

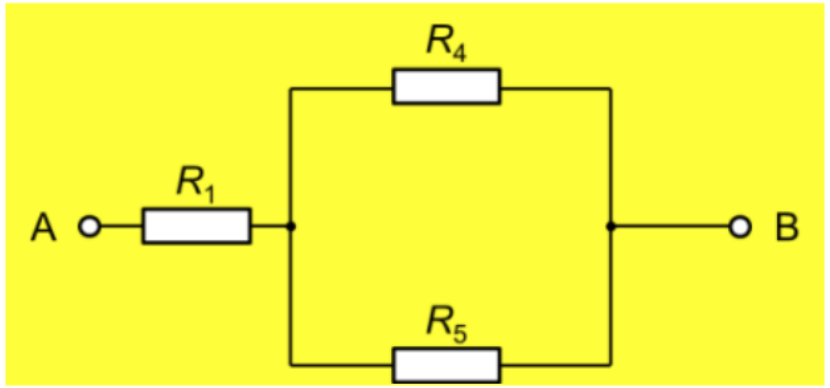
1. Δ to Y

$$R_1 = \frac{R_{12} \cdot R_{31}}{R_{12} + R_{23} + R_{31}} = \frac{1\,\Omega \cdot 2\,\Omega}{1\,\Omega + 3\,\Omega + 2\,\Omega} = 1/3\,\Omega$$

$$R_2 = \frac{1\,\Omega \cdot 3\,\Omega}{1\,\Omega + 3\,\Omega + 2\,\Omega} = 0{,}5\,\Omega\,; \quad R_3 = \frac{2\,\Omega \cdot 3\,\Omega}{1\,\Omega + 3\,\Omega + 2\,\Omega} = 1\,\Omega$$



2. combine



$$R_4 = R_2 + 2\,\Omega = 2{,}5\,\Omega\,; \quad R_5 = R_3 + 1\,\Omega = 2\,\Omega$$

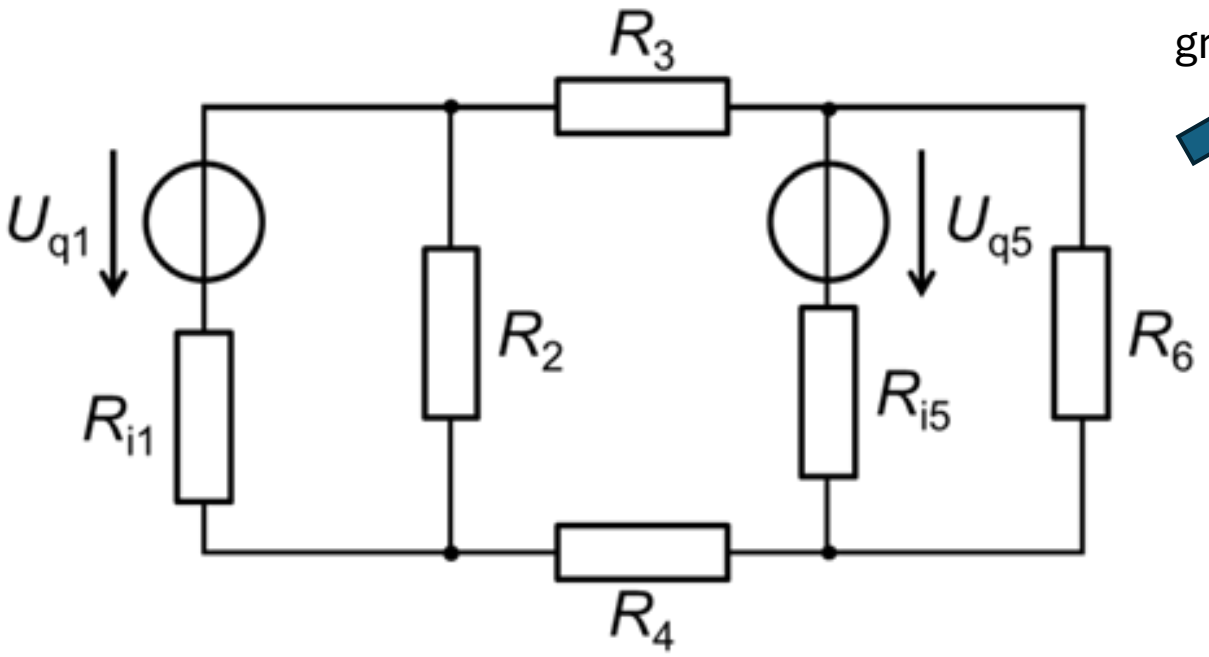
$$R_{AB} = R_1 + \frac{R_4 R_5}{R_4 + R_5} = 1/3\,\Omega + \frac{2{,}5\,\Omega \cdot 2\,\Omega}{2{,}5\,\Omega + 2\,\Omega} = 1{,}44\,\Omega$$

Systematic Network Analysis

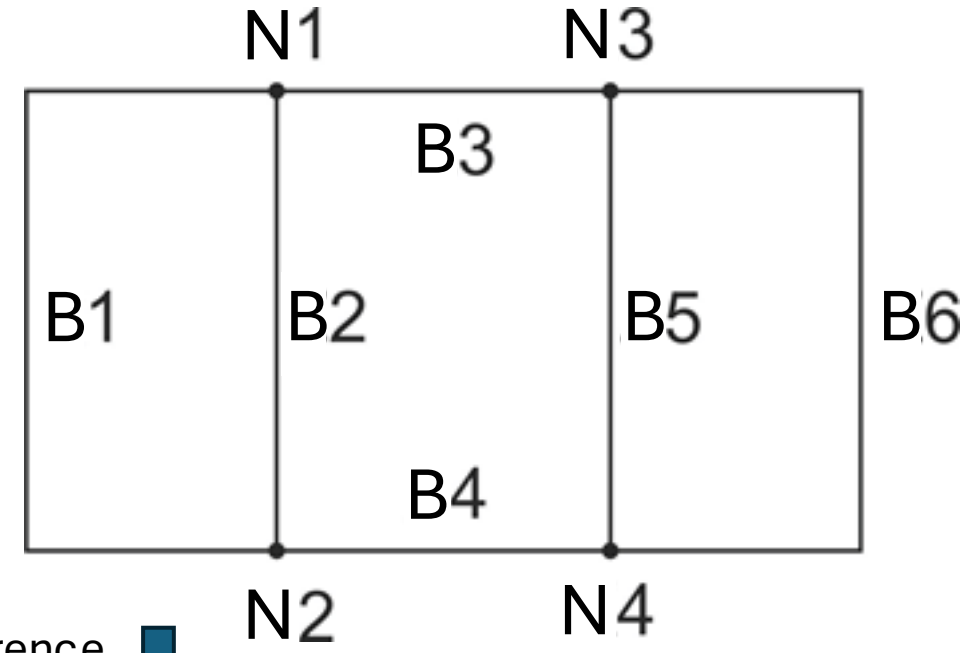
for DC resistive networks

1. represent the network
2. formulate equations: choose analysis method (different heuristics) and apply KCL / KVL / Ohm's law accordingly
3. solve the resulting system of linear equations

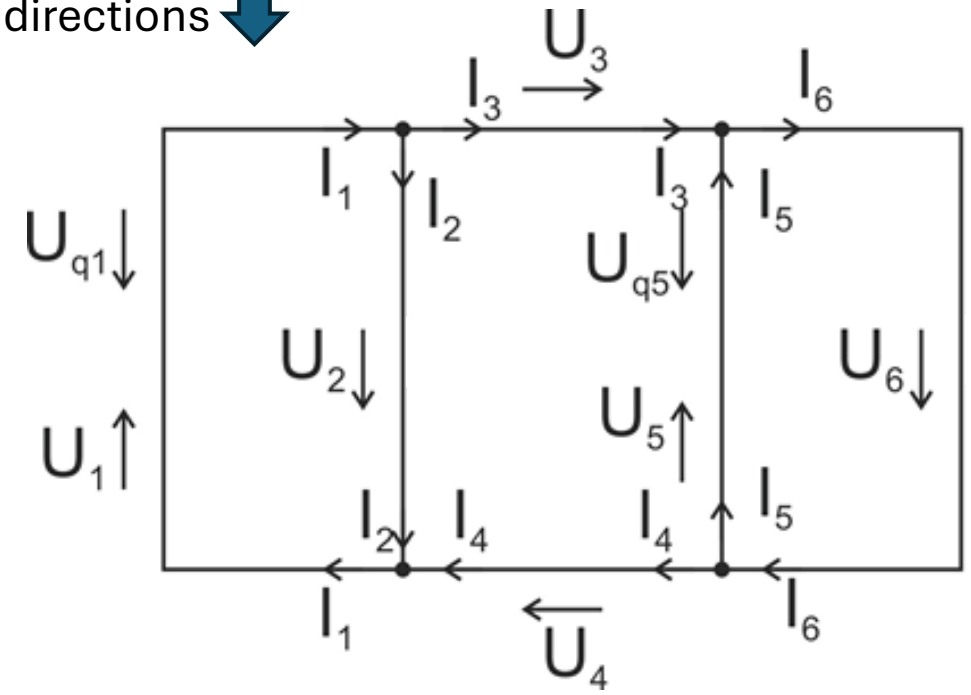
Representation



graph



reference directions



Formulation of Equations

for network with n nodes: $n-1$ linearly independent node equations

here:

- N1: $I_1 - I_2 - I_3 = 0$
- N2: $-I_1 + I_2 + I_4 = 0$
- N3: $I_3 + I_5 - I_6 = 0$

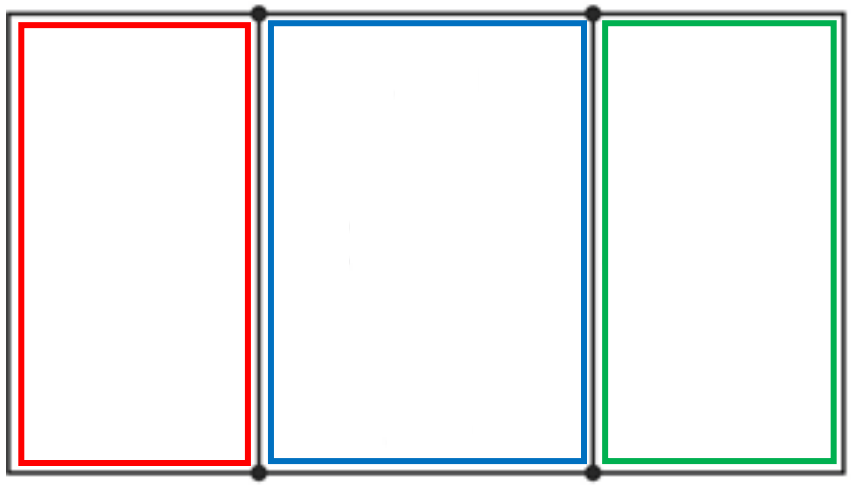
for b branches: $m = b - (n-1)$ linearly independent loop equations

here: $b = 6, n = 4 \Longrightarrow m = 3$ → need to pick 3 out of the 6 loops

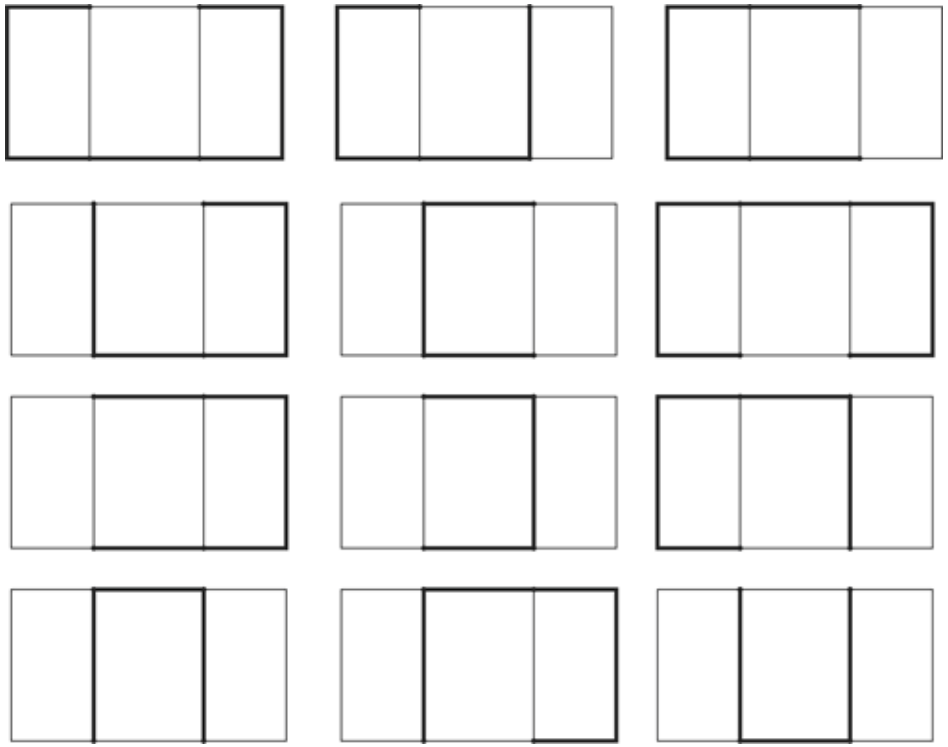
Approach 1: Use Meshes

meshes: smallest possible loops, not enclosing any other loop

here: 3 meshes give 3 linearly independent loop equations



Approach 2: Method of Complete Tree



tree: set of branches connecting all nodes without forming any loop
co-tree: all branches – tree branches

choose one tree:





get fundamental loops (not necessarily meshes) by adding the different co-tree branches back to the tree

M1: $U_1 + U_2 = U_{q1}$

M3: $U_5 + U_6 = U_{q5}$

M4: $U_1 + U_3 + U_4 - U_5 = U_{q1} - U_{q5}$

System of Linear Equations

$$I_1 - I_2 - I_3 = 0$$

$$-I_1 + I_2 + I_4 = 0$$

$$I_3 + I_5 - I_6 = 0$$

$$U_1 + U_2 = U_{q1} \Longrightarrow R_{i1} I_1 + R_2 I_2 = U_{q1}$$

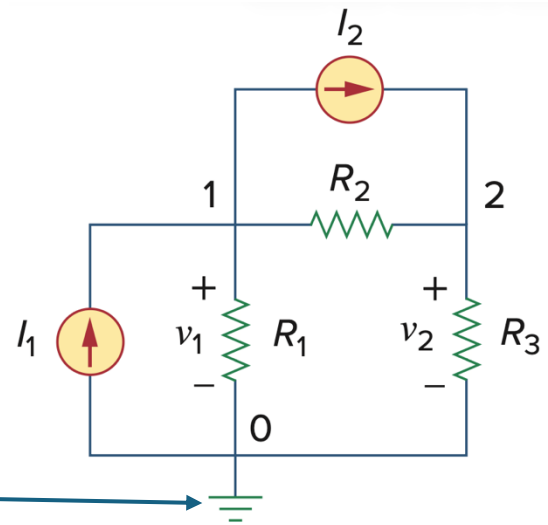
$$U_5 + U_6 = U_{q5} \Longrightarrow R_{i5} I_5 + R_6 I_6 = U_{q5}$$

$$U_1 + U_3 + U_4 - U_5 = U_{q1} - U_{q5} \Longrightarrow R_{i1} I_1 + R_3 I_3 + R_4 I_4 - R_{i5} I_5 = U_{q1} - U_{q5}$$

$$\begin{pmatrix} 1 & -1 & -1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & -1 \\ R_{i1} & R_2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & R_{i5} & R_6 \\ R_{i1} & 0 & R_3 & R_4 & -R_{i5} & 0 \end{pmatrix} \cdot \begin{pmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \\ I_6 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ U_{q1} \\ U_{q5} \\ U_{q1} - U_{q5} \end{pmatrix}$$

Alternative: Nodal Analysis

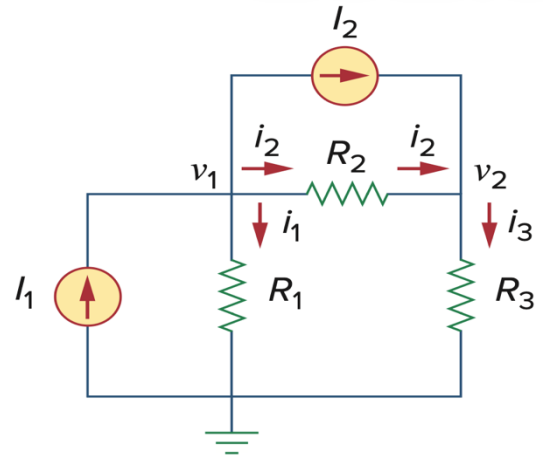
sometimes simpler approach:



- select one node as reference node with an arbitrary potential (typically ground, i.e., $V = 0$)

- apply KCL to $n - 1$ non-reference nodes

redrawn with currents:



here: node1: $I_1 = I_2 + i_1 + i_2$ node2: $I_2 + i_2 = i_3$

- apply Ohm's law to express currents in terms of node voltages:

$$i = \frac{v_{\text{higher}} - v_{\text{lower}}}{R}$$

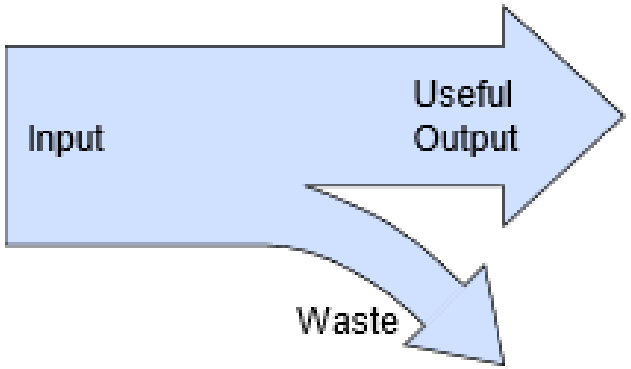
here: $i_1 = \frac{v_1 - 0}{R_1}$ $i_2 = \frac{v_1 - v_2}{R_2}$ $i_3 = \frac{v_2 - 0}{R_3}$

- substitute and solve

Efficiency

useful output

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}}$$

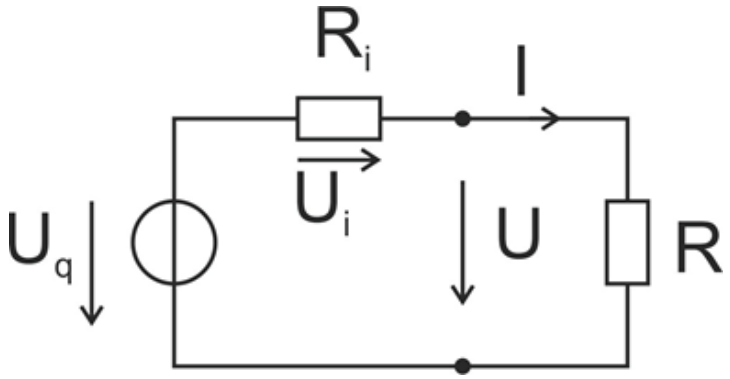


example:

A device draws $200\ \text{W}$ and has a useful output of $120\ \text{W}$.

→ $\eta = 60\%$, i.e., wasting 40% of the input power

Power/Impedance Matching



practical voltage source with resistive load:

$$P_{\text{out}} = U \cdot I; \quad P_{\text{in}} = U_q \cdot I; \quad P_{\text{waste}} = I^2 \cdot R_i$$

$$\eta = \frac{UI}{U_q I} = \frac{U_q \frac{R_L}{R_i + R_L} \frac{U_q}{R_i + R_L}}{U_q \frac{U_q}{R_i + R_L}} = \frac{R_L}{R_i + R_L}$$

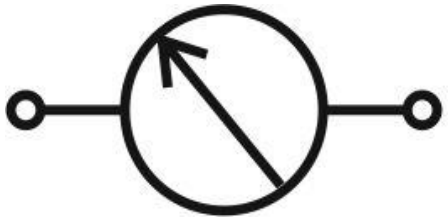
$\eta = 100\%$ at $R_i = 0$

$\eta = 50\%$ at $R_i = R_L$

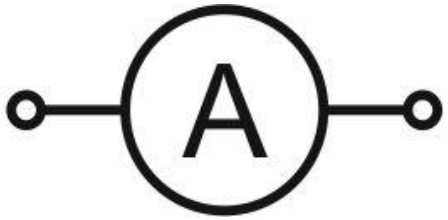
maximum power transfer:

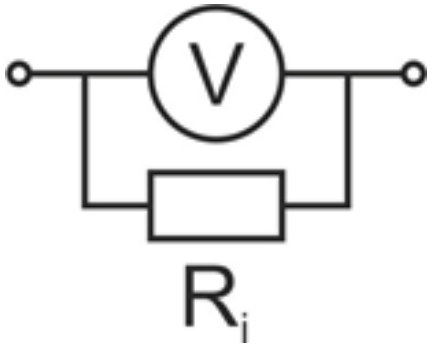
$$\frac{dP_{\text{out}}}{dR_L} = 0 = U_q^2 \frac{(R_i + R_L)^2 - 2R_L(R_i + R_L)}{(R_i + R_L)^4} \Leftrightarrow R_i = R_L$$

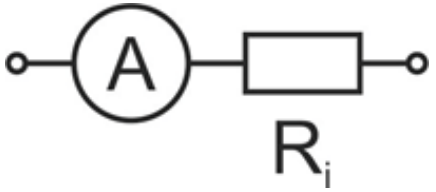
Voltage and Current Measurement











connected in parallel

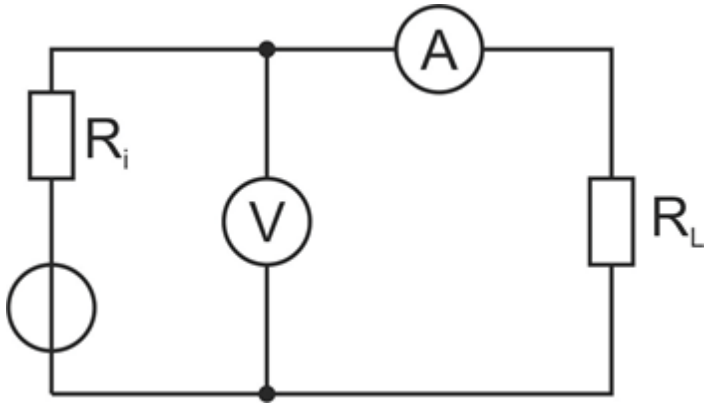
ideal: $R_i = \infty$

connected in series

ideal: $R_i = 0$

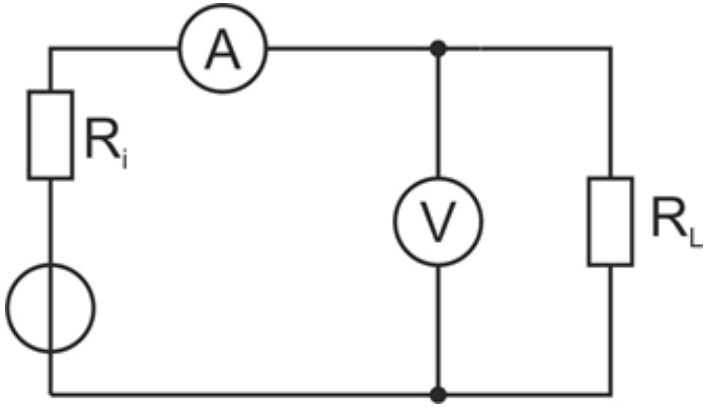
Simultaneous Measurement

for power measurement: need for simultaneous measurement of voltage and current



measurement correct for current

preferable for high R_L



measurement correct for voltage

preferable for low R_L